

# Netflix Content Intelligence

# NETFLIX

## Turning Metadata into Content Strategy Decisions

### Approach

- Exploratory Data Analysis (EDA)
- Feature Engineering
- Machine Learning–driven actions
- Business decision mapping

### Key Framing Points

- Uses *only* Netflix catalog metadata (no external assumptions)
- Focuses on **decision support**, not just visualization
- Designed as a **content lifecycle intelligence system**
- Bridges analytics with **real business levers**
- Scalable across regions and time

 *All insights derived purely from Netflix catalog metadata*



# Dataset Overview (Foundation)



## Dataset Summary

| Metric           | Value            |
|------------------|------------------|
| Total Titles     | 8,810            |
| Content Types    | Movies, TV Shows |
| Time Range       | 2008–2021        |
| Unique Countries | 90+              |
| Genres           | 40+              |
| Ratings          | 14               |



## Insights

- Dataset captures **global expansion phase of Netflix**
- Covers both **pre- and post-originals era**
- Large enough for **portfolio-level strategy**
- Includes cultural, genre, and maturity signals
- Suitable for ML without heavy augmentation



### Why It Matters

Enables **strategic insights**, not anecdotal findings



# Content Type Distribution



## Movies vs TV Shows

| Content Type | Count | Percentage |
|--------------|-------|------------|
| Movies       | 6,100 | 70%        |
| TV Shows     | 2,700 | 30%        |



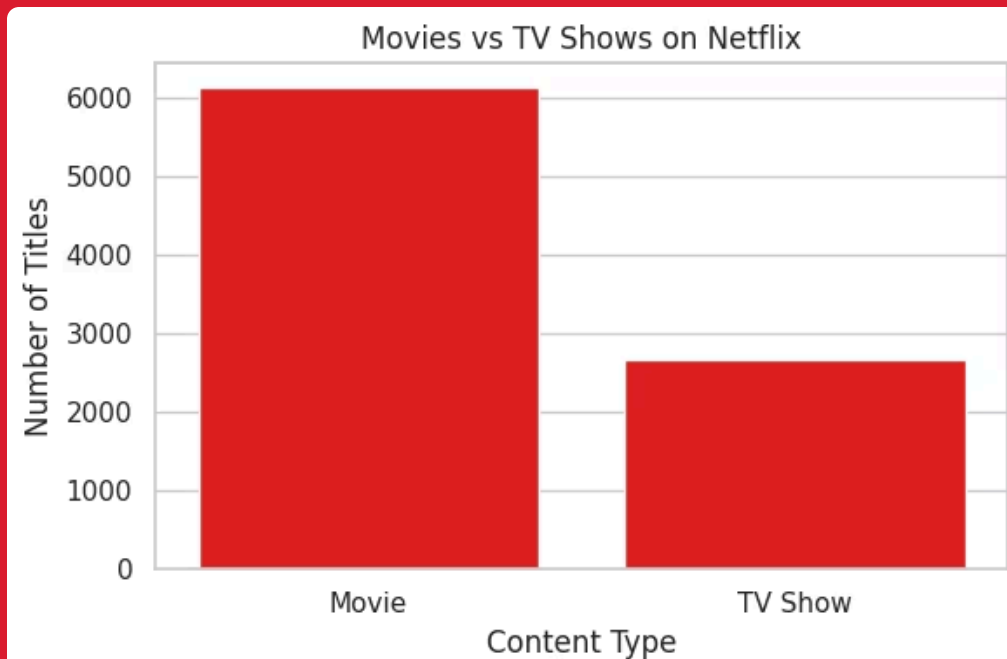
## Insights

- Movies dominate → **volume and discovery strategy**
- TV shows fewer but higher **engagement depth**
- Indicates intentional portfolio balance
- TV shows likely costlier but higher LTV
- Movies reduce "time-to-first-watch" friction



## Actions

- Use movies for **frequent catalog refresh**
- Use TV shows to **anchor long-term retention**
- Promote shows strategically during churn-risk periods





# Content Growth Over Time



## Netflix Content Additions by Year



### Insights

- Sharp acceleration post-2015
- Peak additions around 2018–2019
- Slower growth post-2020 → optimization phase
- TV Shows grew steadily, not explosively
- Reflects shift from expansion to efficiency



### Strategic Meaning

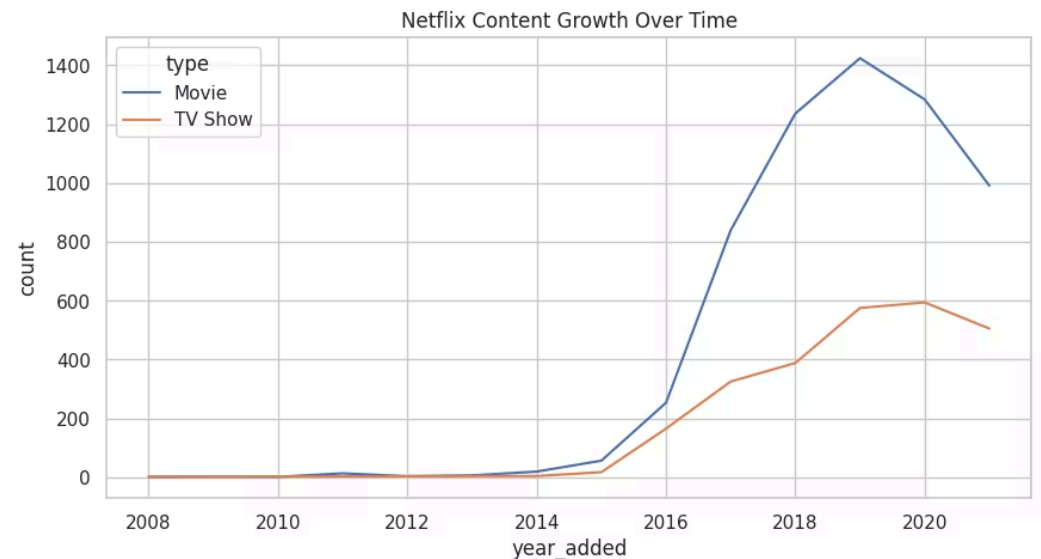
- Netflix transitioned from **scale-first** to **value-first**
- Indicates maturity of the platform



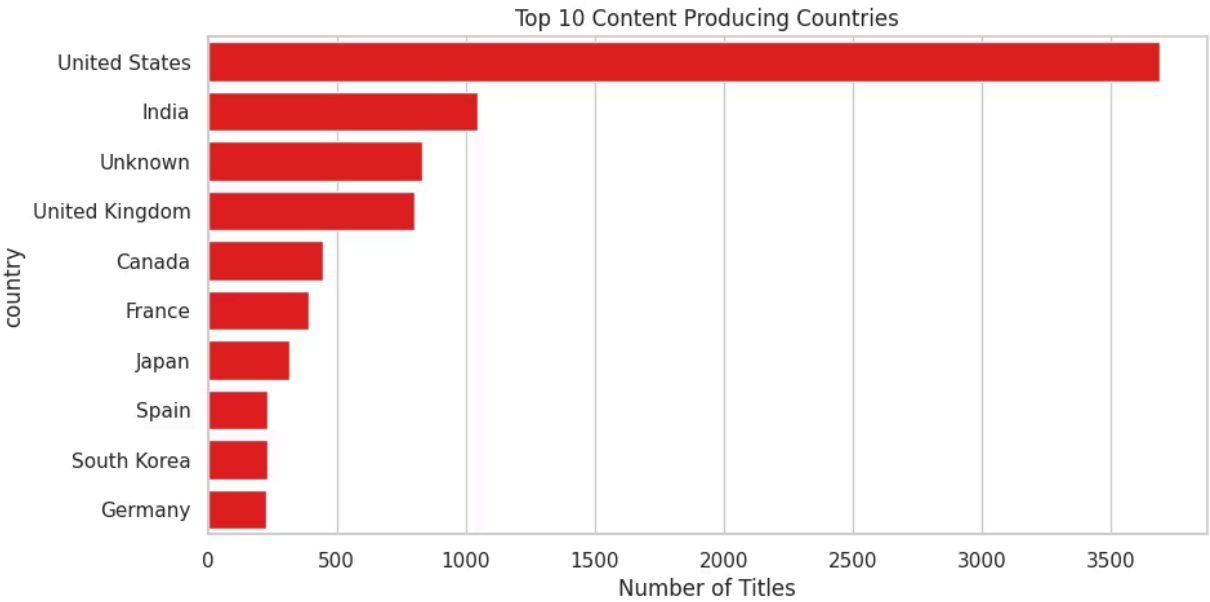
### Actions

Optimize visibility  
of existing  
content

Focus on ROI per  
title, not volume  
alone



# Geographic Content Concentration

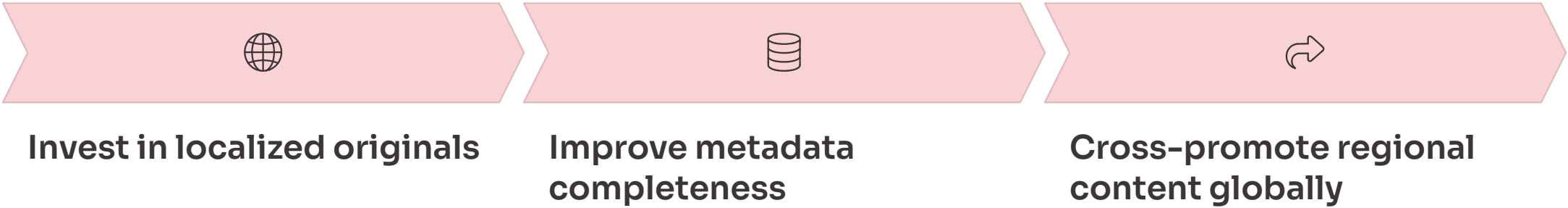


## Top Producing Countries

| Country        | Titles |
|----------------|--------|
| United States  | 3,700  |
| India          | 1,000  |
| United Kingdom | ~800   |

## Insights

- US dominates → content risk concentration
- India is fastest-growing international contributor
- "Unknown" country highlights metadata gaps
- Global diversity increasing post-2016
- Regional hits often become global successes



# Genre Portfolio Analysis

## Top Genres

| Genre                | Relative Presence |
|----------------------|-------------------|
| International Movies | Very High         |
| Dramas               | High              |
| Comedies             | High              |
| Documentaries        | Medium            |

## Insights

- Drama + International = Netflix's core engine
- Documentaries punch above their volume
- Genre clustering shows audience overlap
- Overreliance risks fatigue
- Niche genres drive loyalty, not volume

## Actions

Maintain genre balance

Fund niche genres for differentiation

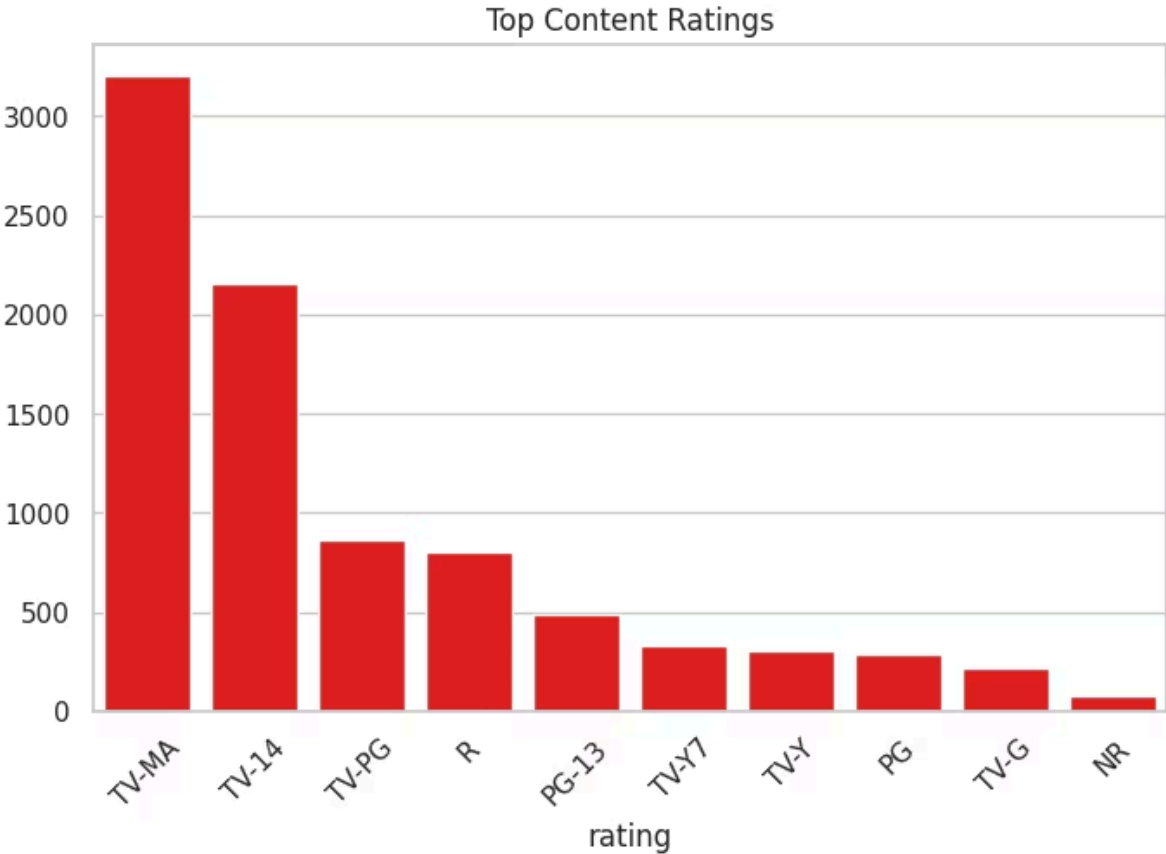
Rotate promotional focus periodically



# Content Rating Strategy



## Rating Distribution



### TV-MA

Adult engagement

### TV-14

Mass-market

### PG / TV-Y

Family



## Insights

- Adult-focused content dominates
- Kids content smaller but sticky
- Ratings act as **implicit segmentation**
- High-TV-MA implies bold content positioning
- Lower competition in kids segment



## Actions

- Expand kids & family content
- Bundle family content for retention
- Target households, not just individuals



# Content Age Analysis



Content Age (Year Added – Release Year)



## Insights

- Median content age = **1 year**
- Netflix prioritizes fresh releases
- Long tail exists up to 90+ years
- Older content survives only if evergreen
- Age is a strong decision variable



## Actions

01

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**Automate age-based promotion rules**

02

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**Identify evergreen exceptions**

03

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**Reduce spend on aging low-value titles**



# Evergreen Content Identification

## Evergreen Genres (Normalized)

| Genre                           | Evergreen Score |
|---------------------------------|-----------------|
| Documentaries                   | High            |
| Kids' TV                        | Very High       |
| Stand-Up Comedy                 | High            |
| Dramas and International Movies | High            |

### Insights

- These genres age slowly
- Cultural relevance remains stable
- Low marketing dependency
- High repeat-watch probability
- Ideal for long-term retention

### Actions

- Lock long-term licenses
- Feature in "comfort watch" sections
- Promote during low-content cycles



# Feature Engineering Logic



## Engineered Features

### Content Age

**Business Meaning:** Freshness

### Genre Count

**Business Meaning:** Breadth

### Rating

**Business Meaning:** Audience

### Type

**Business Meaning:** Engagement depth



## Insights

- Features map directly to business logic
- No black-box variables
- Improves model explainability
- Allows executive trust
- Enables action mapping

### Value

**Makes ML outputs interpretable for leadership**